

Combining quantum circuits in Qiskit

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▲ In Qiskit the instruction to add two quantum circuits

0 $qc1 + qc2$

produces the following error

"unsupported operand type(s) for +: 'QuantumCircuit' and 'QuantumCircuit'"

I'm guessing my library must be missing something but as I'm new to Qiskit and Python I'm not sure what?

I was expecting qc2 to be appended to qc1.

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asked May 1, 2023 at 1:45

 [user21783727](#)
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1 Answer

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The `+` operator is no longer supported by QuantumCircuit. From [Qiskit Terra 0.23 release notes](#) (Upgrade section).

1 The `QuantumCircuit` methods `combine` and `extend` have been removed following their deprecation in Qiskit Terra 0.17.0. This was done because these functions were simply less powerful versions of `QuantumCircuit.compose()`, which should be used instead.

The removal of `extend` also means that the `+` and `+=` operators are no longer defined for `QuantumCircuit`. Instead, you can use the `&` and `&=` operators respectively, which use `QuantumCircuit.compose()`.

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answered May 1, 2023 at 12:16

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